Muthaiya Thavamani

AI-Native Engineer | Technical Architect

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AI-Native Engineer and Technical Architect with 11 years of experience, operating as a Forward Deployed Engineer - embedding with teams to take Generative AI from discovery to production. I own deployments end to end: agentic systems, RAG, and copilots that take real actions, backed by strong data engineering and predictive ML. I've shipped products from architecture through rollout across multiple early-stage startups and enterprise clients.

Skills

AI / ML Engineering	- Generative AI & LLM apps: foundation models (OpenAI, Anthropic Claude, LLaMA), RAG, prompt engineering, fine-tuning, embeddings, vector databases - Agents: multi-agent, LangChain, LangGraph, MCP, tool/function calling, copilots; LLM evaluation, guardrails, Responsible AI, HITL; TensorFlow, forecasting, anomaly detection, computer vision
Data Engineering	- Serverless ETL & delta pipelines, data validation, governance & compliance - Databricks Certified Professional, medallion architecture, PySpark, Delta Lake, Teradata, Airflow
Blockchain & Web3	- Solidity, OpenZeppelin, HardHat, EVM; Decentralized Identity (Privado ID) - Polygon CDK, Avalanche Subnet, Cosmos SDK, Tezos (SmartPy), Hyperledger Fabric
Languages, Frameworks & Data	- Golang, Python, JS / TS (NodeJS), Solidity, Java; - Flutter (Dart) & React Native; - FastAPI, ExpressJS, Hasura (GraphQL) - PostgreSQL, MySQL, MongoDB, DynamoDB, Cloud Firestore
Cloud & Infra	- AWS, GCP & Azure - Serverless & microservice deployment, Docker

Professional Experience

Lead AI Engineer, Contract Sep 2025 - Present

- Forward-deployed AI delivery - embedded with clients to ship Generative-AI products from discovery to production.

- **AIDLC - AI-Ops Platform:** a LangChain / LangGraph multi-agent engine that automates the path from requirement to delivery, with prompt engineering and LLM evaluation built into the loop.
- **Governed Asset Agents (Enterprise Asset Management):** a multi-agent system that predicts, analyses, and acts on asset operations - combining anomaly detection with LLM reasoning, and wrapping autonomous actions in guardrails, Responsible-AI checks, and human-in-the-loop approval so every action stays auditable.
- **AI Copilot over MCP:** connects to any system through a secured Model Context Protocol link, using RAG over enterprise data plus tool / function calling to turn scattered tools into one conversational interface, with evaluation keeping responses grounded.
- **Predictive HVAC Engine:** an agentic copilot and predictive engine for healthcare HVAC, applying time-series forecasting and anomaly detection on live telemetry to flag failures before they happen.
- **LLM Document Extraction:** a parser and content-extraction engine using embeddings, vector search, and RAG to turn unstructured enterprise documents into reliable structured data.

Technologies: Python, Generative AI, LLMs (OpenAI, Claude, Hugging Face, LLaMA), LangChain, LangGraph, MCP, RAG, Vector DBs, FastAPI, Databricks, Teradata, DynamoDB, ReactJS, AWS.

Lead Data Engineer, Greenmen Consulting Oct 2024 - Aug 2025

- Led end-to-end data migration from S3, SQL, Databricks, and Teradata into Salesforce CRM, embedded with data owners to enforce data quality, governance, and compliance.
- Built a serverless system automating delta (incremental) loads over millions of records, and a Databricks medallion architecture (Bronze/Silver/Gold) feeding curated data to the PowerBI team.
- **BRD Engine** - Implemented this multi-agent system (LangChain / LangGraph) from ideation through production.

Technologies: Python, Databricks, PySpark, Delta Lake, Multi-Agent AI, Teradata, DynamoDB, AWS.

Product Development Head, Unibase Labs June 2022 - Sep 2024

- Partnered with founders to take Web3 platforms from idea to production, owning architecture, decentralized identity (DID), and L1 development on Avalanche Subnet and Polygon CDK.
- **Turf Network** - led an esports GameFi ecosystem end to end; built a Privado ID identity system publishing 400K+ DIDs and Solidity contracts for the Esports Engine, token sale, and node sale.

- **Instaraise (Tezos Launchpad)** - re-architected the IDO platform as serverless, leading backend services and the entire AWS infrastructure.
  Technologies: Golang, Python, Solidity, Polygon CDK, Privado ID, Tezos, Cosmos SDK.

Lead Blockchain Engineer, Coinfantasy Dec 2021 - June 2022

- Founding engineer - built smart contracts (game, stake/unstake, ERC20) and a Golang game scheduler handling high volumes of parallel events.
- Integrated Biconomy gasless transactions, Lossless protocol, and providers (Magic.link, Transak, CoinGecko, CoinMarketCap); mentored and helped build the core team.
  Technologies: Golang, NodeJS, Solidity, HardHat, MySQL, MongoDB, ReactJS, AWS.,

Backend Security Engineer, Securex AI Oct 2021 - Mar 2022

- Core Golang backend engineer - integrated cloud-security platforms (Veracode, Dome9, Qualys, Azure Defender) and AWS/GCP asset connectors via their official SDKs.
- Built a vulnerability service pulling CVE data and remediations from the NVD.
  Technologies: Golang, Python, AWS, GCP, Azure, Hasura, Postgres.,

Lead Member Technical Staff, Imstrong Dec 2019 - Nov 2021

- Built an ML fitness product using TensorFlow PoseNet - real-time pose estimation (computer vision) scoring yoga and exercise posture accuracy with live form feedback.
- Backend engineer for the video-streaming platform: Golang microservices, a self-built serverless live-streaming system (nginx-rtmp + Docker), VOD tooling, and AWS/GCP infrastructure.
  Technologies: Python, TensorFlow (PoseNet / Computer Vision), Golang, NodeJS, Firestore, nginx-rtmp, AWS, GCP.

Software Engineer, Sysvine Technologies Dec 2015 - Nov 2019

- Peloton Blockchain - ERC20 chaincode in Go on Hyperledger Fabric (Fabric Node SDK).
- Broadridge (Dojima Trade & Clear) - order/market-data services over FIX/FIXML and the NASDAQ Genium API.
- Elsevier (GeoFacets) - content & metadata extraction, OCR, and geotagging pipelines for map-based discovery.
  Technologies: GoLang, Java, NodeJS, Python, Blockchain (Hyperledger), Websocket, IBM MQ